



SARDAR PATEL UNIVERSITY
B.C.A.(5thSEM) – EXAMINATION OCT-2023
US05DBCA56: COMPUTER GRAPHICS

Date: 30-10-2023(Monday)

Time: 10:00 AM to 01:00 PM

NOTE: Right hand figure indicates the marks of each question.

Total Marks: [70]

Q. 1 Multiple Choice Questions (One marks each)**[10]**

- 1 _____ produce illustrations for reports for use with projectors.
 (A) Computer Art (B) Image Processing
 (C) Entertainment (D) Presentation Graphics
- 2 _____ is also known as calligraphic display.
 (A) Random scan display (B) Raster Scan display
 (C) Proper Display (D) None of the above
- 3 Beam Penetration Method displays maximum _____ colors.
 (A) 8 (B) 4
 (C) 16 (D) 32
- 4 _____ type has small lines or accents at the end of the main character.
 (A) Symbols (B) Sans-serif
 (C) Verdana (D) Serif
- 5 _____ obtained by adding a filled semicircle to each butt cap.
 (A) Round cap (B) Circle cap
 (C) Projecting Square Cap (D) Semi-circle cap
- 6 Boundary fill algorithm can be applied as _____.
 (A) 4-connected approach (B) 8-connected approach
 (C) 12-connected approach (D) Both (A) and (B)
- 7 Rotation point is also known as _____.
 (A) Rotation angle (B) Rotation axis
 (C) Pivot point (D) Translation vector
- 8 A scaling transformation alters the _____ of an object.
 (A) Size (B) Orientation
 (C) Shape (D) None of these
- 9 _____ mode application initiates data entry.
 (A) Request Mode (B) Event Mode
 (C) Sample Mode (D) None of above
- 10 3D Graphics include _____.
 (A) Height (B) Depth
 (C) Width (D) All of the above

Q. 2 Short Questions. (Attempt any ten) [20]

- 1 Define the terms: (1) Bitmap, (2) Pixel
- 2 Give full form of CRT and List all the primary components of CRT.
- 3 Describe the working of Joystick in brief.
- 4 List joins used at the end of line and Explain any one.
- 5 Differentiate: Bitmap font and Outline font.
- 6 What is Inside-Outside Test.? List methods used in Inside-Outside Test.
- 7 Define the term Scaling. List out types of Scaling.
- 8 Explain the term clipping and list out all types of clipping.
- 9 Define the terms: (1) Reflection, (2) Viewport.
- 10 Describe Virtual Reality Environment in brief.
- 11 What is Depth Cueing?
- 12 Define the terms: (1) Application Icons, (2) Control Icons.

Q. 3 A] Describe working of CRT in detail. [05]

B] Write note on input devices Keyboard and Mouse. [05]

OR

Q. 3 A] Differentiate between Raster scan displays and Random scan displays. [05]

B] Write note on LCD and LED. [05]

Q. 4 A] Explain Bresenham line drawing algorithm with example. [05]

B] Explain Odd-Even rule with example. [05]

OR

Q. 4 A] Explain DDA algorithm with steps and example. [05]

B] Write a note on Line attributes. [05]

Q. 5 A] Explain Sutherland Hodgeman polygon clipping algorithm in detail. [05]

B] What is geometric transformation? List all 2-D geometric transformation and explain any one in detail. [05]

OR

Q. 5 A] What is text clipping? List out all types and explain it with example. [05]

B] Explain Cohen Sutherland Line Clipping Algorithm in detail. [05]

Q. 6 A] Explain General considerations in structuring a User Dialogue in detail. [05]

B] Explain purpose of graphical input functions and explain different input modes. [05]

OR

Q. 6 A] List and explain interactive picture-construction techniques. [05]

B] Explain Polygon surfaces in detail. [05]